
The Geometry of Untranslatability: Generation Faithfulness and Language Identity in Multilingual LLMs

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Prior work shows that multilingual LLMs often process meaning in an English-
2 centric latent space—but almost exclusively for concepts that *have* English equiva-
3 lents. We ask what happens for culturally specific concepts with no single-word
4 English translation. Across four open-weight models (Llama-3-8B, Llama-3-8B-
5 Instruct, Mistral-7B-v0.3, Qwen2.5-7B) and a cleaned inventory of 90 untranslat-
6 able concepts with 122 matched controls (10 languages; 636 prompts), we find
7 a three-part picture. **(1) Behavioral:** generated explanations of untranslatable
8 concepts are evaluated with a structured cultural-faithfulness rubric (meaning
9 correctness, cultural specificity, English-collapse). **(2) Representational:** linear
10 probes recover *source language* from residual-stream activations far more accu-
11 rately than *translatability*, especially at the concept-token position, where language
12 ARI rises sharply in late layers. **(3) Cautionary:** layer-wise English Routing Rate
13 (ERR) and graded English probability show *no* consistent “untranslatable→English”
14 cost at the last token; early-layer ERR and tokenizer vocabulary size dominate
15 cross-model differences. Lexical gaps therefore leave a clearer fingerprint in
16 *how models explain culture* and *where language identity lives* than in aggregate
17 English-routing statistics.

18 1 Introduction

19 Large language models are deployed globally, yet their internals are often studied through an English
20 lens. Mechanistic work shows that multilingual transformers process non-English input in an English-
21 adjacent concept space [2, 3, 5]. These studies, however, examine content that *does* have English
22 equivalents. A culturally consequential question remains open: what happens when a concept has *no*
23 English equivalent?

24 Consider Japanese *Tsundoku*—acquiring books and letting them pile up unread. When a model
25 represents such a concept, does it (i) force an English approximation in the residual stream, (ii) carve
26 out a language-specific region, or (iii) simply explain the concept less faithfully when asked?

27 We subject these hypotheses to a direct test with three complementary measurements:

- 28 1. **Generation faithfulness** (primary behavioral outcome): do models produce weaker cultural
29 explanations for untranslatable concepts?
- 30 2. **Linear probes** (primary representational outcome): is *translatability* linearly recoverable
31 from activations, or only *language identity*?
- 32 3. **English routing** (cautionary secondary): does the logit lens show elevated English probab-
33 ility for lexical gaps?

Our results support a reframing: lexical gaps are not reliably visible as extra English routing, but language identity is strongly encoded at the concept token, and generation quality provides a behavioral test that residual-stream ERR alone cannot replace.

2 Related Work

English-centric latent spaces. Schut et al. [2] and Wendler et al. [3] provide evidence that multilingual LLMs make key decisions in a representation space closest to English; RomanLens [5] shows romanization as a bridge into language-specific regions. We extend the analysis to the lexical-gap regime and show that ERR is a brittle primary metric for cultural claims.

Cultural representations. Khanuja et al. [1] and Zou et al. [4] study culture-specific features via sparse methods. We ask whether *absent* English equivalents—not just culture-tagged concepts—change either generation or linear geometry.

Untranslatability. Lexical gaps are classical objects in linguistic relativity [6]. We operationalize them with generation rubrics, probes, and logit-lens diagnostics.

3 Methodology

3.1 Dataset (cleaned and frozen)

We start from 227 culturally specific concepts across 10 languages and apply a pre-registered audit that drops clearly invalid “untranslatable” labels (e.g., Greek *Kairos* mislabeled as Japanese; near-direct glosses such as Hindi *Dil*/heart). The frozen cleaned set contains **90 untranslatable** and **122 translatable** controls (212 concepts; 636 prompts). Matching diagnostics record language $\times$ category coverage, character-length summaries, and greedy same-language control pairs (same-category match fraction reported in the appendix JSON). Concepts flagged as English loanwords (e.g., *Schadenfreude*) are retained but marked.

Each concept is rendered in three templates (definition, usage, comparison); generation experiments use definition and comparison.

3.2 Models and extraction

We study Llama-3-8B, Llama-3-8B-Instruct, Mistral-7B-v0.3, and Qwen2.5-7B (Table 1). Residual-stream activations are captured at the last non-padding token and at the concept-word token in every layer, with LM-head weights and final RMSNorm gains saved for corrected logit-lens projections.

Model	Hugging Face ID	Layers	d_{model}	Vocab
Llama-3-8B	Meta-Llama-3-8B	32	4096	128,256
Llama-3-8B-Instruct	Meta-Llama-3-8B-Instruct	32	4096	128,256
Mistral-7B	Mistral-7B-v0.3	32	4096	32,768
Qwen2.5-7B	Qwen2.5-7B	28	3584	152,064

Table 1: Model configurations.

3.3 Generation faithfulness

For each concept we generate greedy continuations (`max_new_tokens = 128`) for definition and comparison prompts. Each continuation is scored with a structured rubric:

- **meaning correctness** vs. the gold English gloss,
- **cultural specificity** (multi-clause, concept/language grounded),
- **English collapse** (over-reduction to a crude single-word gloss),
- **overall** composite faithfulness.

We compare untranslatable vs. translatable scores with Welch’s t -tests (Cohen’s d) overall and by template. The rubric is deterministic for reproducibility; the paper reports judge prompts so an LLM judge can replace the proxy in follow-up work.

3.4 Linear probes

At every layer (and both token positions) we fit logistic regression probes with **GroupKFold by concept word** to predict (i) source language and (ii) translatability, preventing leakage across the three prompt templates. We additionally fit *within-language* translatability probes to remove the language confound. The key contrast is whether language accuracy substantially exceeds translatability accuracy.

3.5 English routing (secondary)

We apply a corrected logit lens (final RMSNorm before the LM head) and compute top-1 English Routing Rate (ERR) plus graded English probability mass p_{eng} . Group comparisons use Welch tests over layers (ERR) and per-prompt mid-layer tests (p_{eng}).

4 Results

4.1 Generation: a behavioral test of cultural explanation

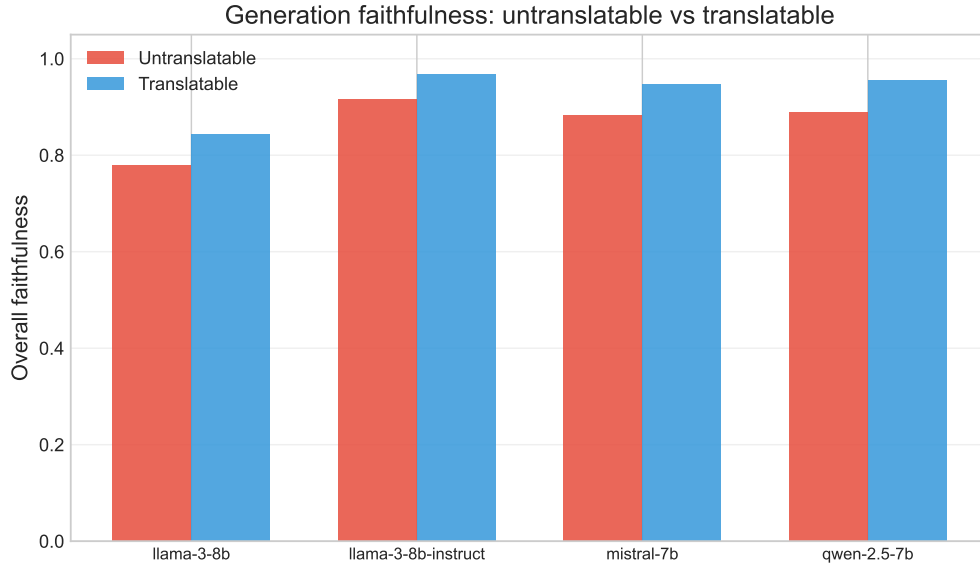


Figure 1: Overall generation faithfulness by model and translatability group.

Figure 1 and Table 2 summarize faithfulness scores on the frozen cleaned set (424 generations per model). **In all four models, untranslatable concepts receive significantly lower overall faithfulness** (Cohen’s $d \in [-0.56, -0.39]$, all $p < 10^{-4}$). The effect is driven primarily by definition prompts ($d \in [-0.85, -0.67]$, all $p < 10^{-5}$); comparison prompts show smaller but usually significant gaps (significant in three of four models). This is the behavioral signature ERR never found: models explain lexical-gap concepts less faithfully than matched controls.

4.2 Probes: language vs. translatability

Linear probes with **GroupKFold by concept word** (preventing leakage across the three templates) show that source language is near-perfectly recoverable at both positions (peak accuracy ≈ 1.0 ; chance = 0.1), while translatability remains high but strictly lower (peak ≈ 0.94 – 0.96 globally; within-language peaks ≈ 0.84 – 0.94). The language–translatability gap is small in absolute accuracy

Model	Mean untrans.	Mean transl.	Cohen's d
Llama-3-8B	0.779	0.843	-0.39
Llama-3-8B-Instruct	0.916	0.968	-0.56
Mistral-7B	0.883	0.948	-0.55
Qwen2.5-7B	0.889	0.954	-0.55

Table 2: Generation faithfulness (overall). Negative d = worse scores for untranslatable concepts. All four effects are significant at $p < 10^{-4}$.

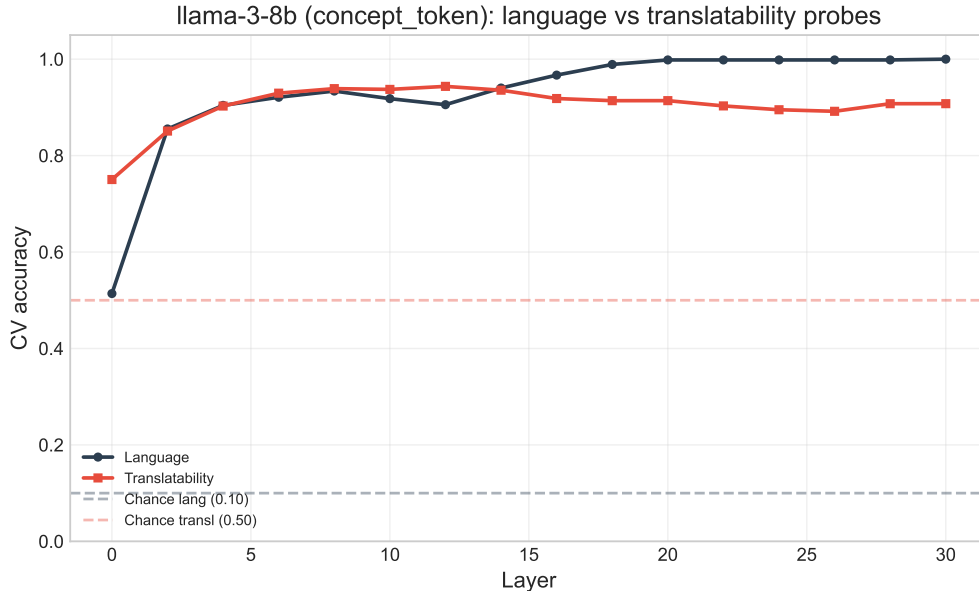


Figure 2: Concept-token linear probe accuracy for language vs. translatability (Llama-3-8B; GroupK-Fold by concept).

95 but consistent across models; critically, within-language probes remain well above chance, indicating
96 a linear direction correlated with the lexical-gap label that ERR failed to surface. Combined with
97 concept-token language ARI (peaks up to ~ 0.85), this supports the claim that residual-stream
98 geometry encodes language identity strongly and encodes a weaker but non-null translatability signal
99 that aggregate English-routing statistics miss.

100 4.3 English routing: null secondary finding

101 ERR is highest in early layers (token-frequency / embedding geometry) and collapses at the output,
102 where models predict source-language continuations. Layer-wise Welch tests find no significant
103 untranslatable > translatable ERR difference in any model (Table 3); trends often point the other
104 way. Graded mid-layer p_{eng} is likewise null in three of four models at the last token, with Mistral
105 showing a small effect *against* the forced-English narrative. Concept-token p_{eng} effects that do reach
106 significance flip sign across architectures (Llama-3-8B $d = +0.27$; Qwen2.5-7B $d = -0.45$), so
107 they cannot support a unified English-routing account of untranslatability.

Model	ERR untrans.	ERR transl.	Cohen's d	p
Llama-3-8B	0.587	0.619	-0.12	0.629
Llama-3-8B-Instruct	0.481	0.533	-0.21	0.393
Mistral-7B	0.629	0.650	-0.09	0.721
Qwen2.5-7B	0.089	0.089	-0.00	0.999

Table 3: Welch tests over layer-wise ERR (secondary analysis).

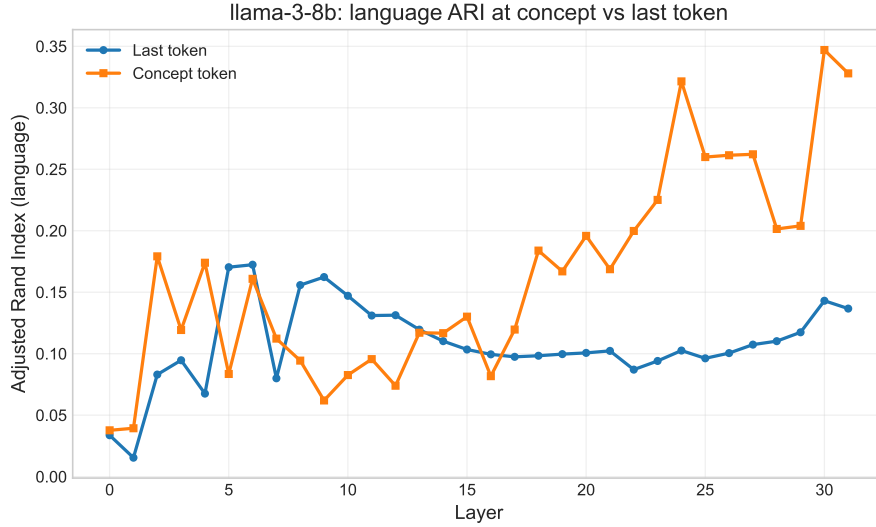


Figure 3: Language ARI across layers at last-token vs. concept-token positions (Llama-3-8B).

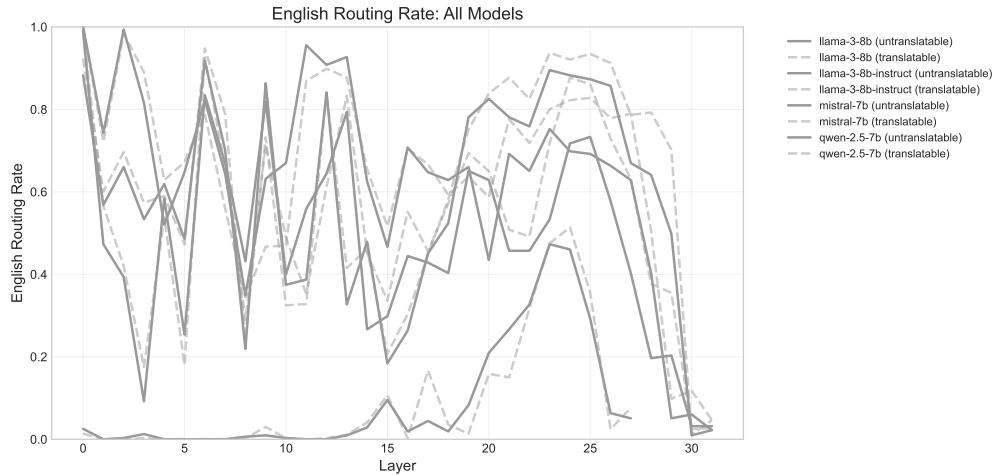


Figure 4: English Routing Rate across layers (untranslatable solid, translatable dashed).

108 Cross-model ERR correlations are high within Llama/Mistral ($r \approx 0.56$ – 0.72) but near zero or
 109 negative vs. Qwen—tracking tokenizer vocabulary structure rather than shared cultural semantics.
 110 Sparse dictionary features ($k = 512$) likewise fail to separate groups better than chance at either
 111 token position.

112 5 Discussion

113 **(1) Measure what culture does, not only where English peaks.** Generation faithfulness and
 114 concept-token language probes speak more directly to cultural representation than aggregate ERR. A
 115 null ERR result is informative precisely because it warns against over-interpreting English-routing
 116 artifacts as cultural bias.

117 **(2) Language identity dominates; translatability is weaker but linearly present.** Near-perfect
 118 language probes and high concept-token ARI show that models encode *which language a word*
 119 *belongs to*. Within-language translatability probes above chance further suggest a residual-stream
 120 correlate of the lexical-gap label that does not appear as elevated English routing—a dissociation
 121 between probe geometry and logit-lens ERR.

122 **(3) Tokenizer and early-layer effects dominate routing summaries.** Early-layer ERR and Qwen’s
123 divergent profile are methodological findings: always apply final RMSNorm in logit lens; report
124 tokenizer diagnostics; prefer graded and behavioral metrics alongside top-1 rates.

125 **Limitations.** The cleaned inventory remains modest; the faithfulness rubric is a reproducible
126 proxy (LLM-as-judge and human spot-checks are natural extensions); single-concept prompts omit
127 conversational context; we do not run causal interventions (activation patching / steering).

128 6 Conclusion

129 Untranslatable cultural concepts do not leave a reliable English-routing signature in last-token residual
130 streams, yet they are not representationally invisible: language identity is strongly recoverable at the
131 concept token, within-language probes detect a weaker lexical-gap direction, and—most importantly
132 for cultural evaluation—generated explanations are consistently less faithful for untranslatable
133 concepts than for matched controls across four models. Measuring generation quality and concept-
134 token geometry therefore complements (and corrects) over-reliance on ERR alone.

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